



PicoMuon Detector Setting up a Raspberry Pi 4/5

Pico^μ

<https://ukraa.com/>

The UK Radio Astronomy Association

A charitable incorporated organisation

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Set-up instructions

These instructions cover setting up your Raspberry Pi4 or Raspberry Pi 5 with a suitable operating system and ability to be remotely access via VNC.

It is recommended that you use a Raspberry Pi 4/5 with at least 2GB memory.

When using the Raspberry Pi Imager, remember to write down the **hostname** you use – this is used to view the web page with the results from your UKRAA PicoMuon detector.

Please keep the username as **pi** when using OS customisation within the Raspberry Pi Imager.

The instructions for accessing your Raspberry Pi heedlessly assume use of Windows PC and that the following two applications are preinstalled on that PC.

- PuTTY
- RealVNC Viewer

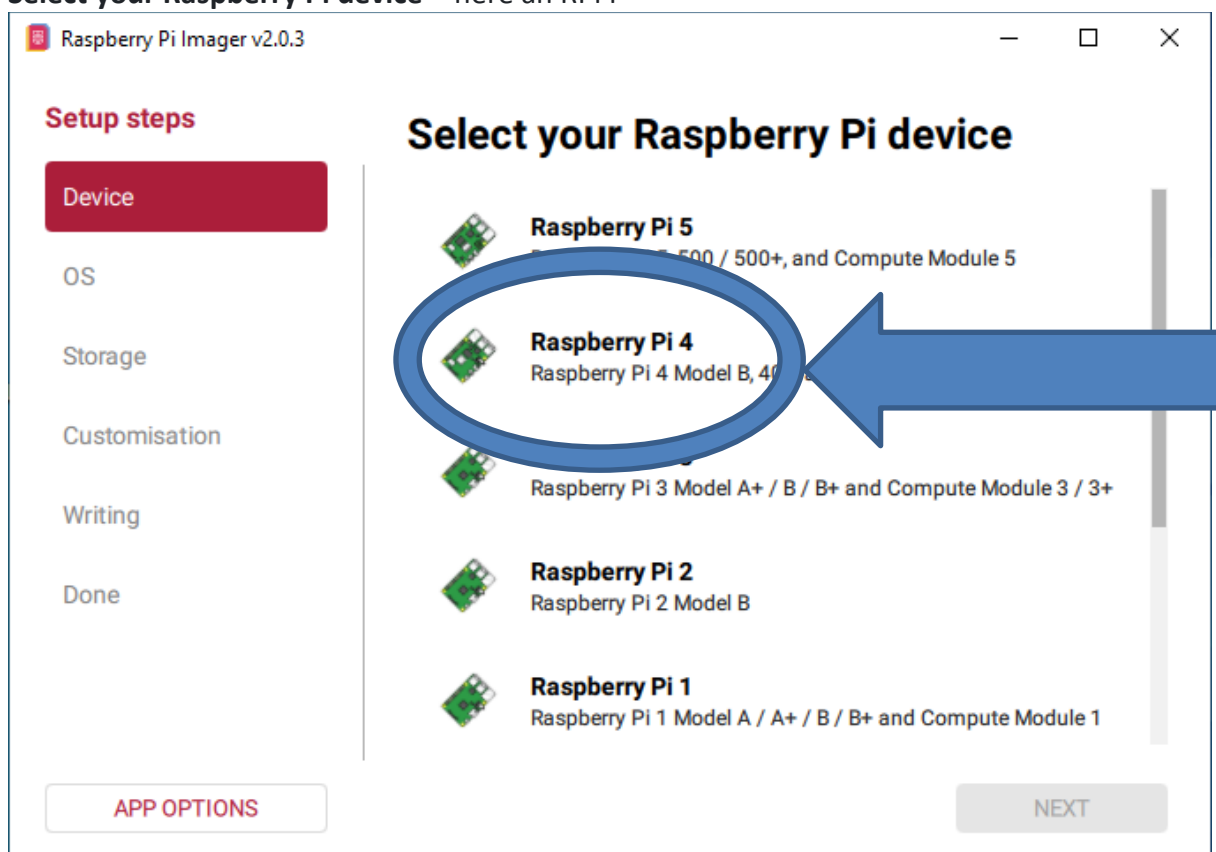
No information on installing these two applications is given in this text – there are numerous guides available on the internet.

Raspberry Pi OS

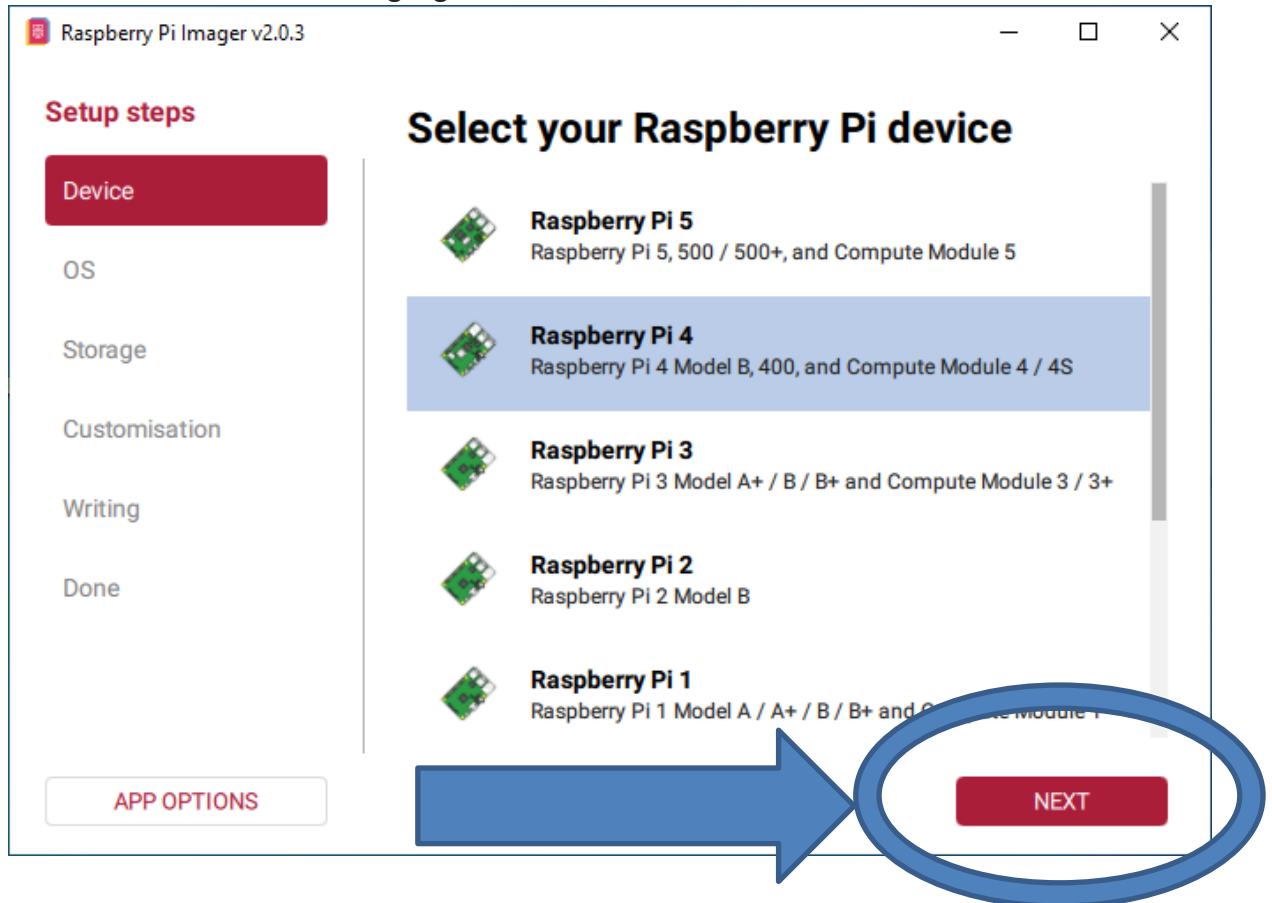
1. Download [Raspberry Pi Imager](#) for your operating system.



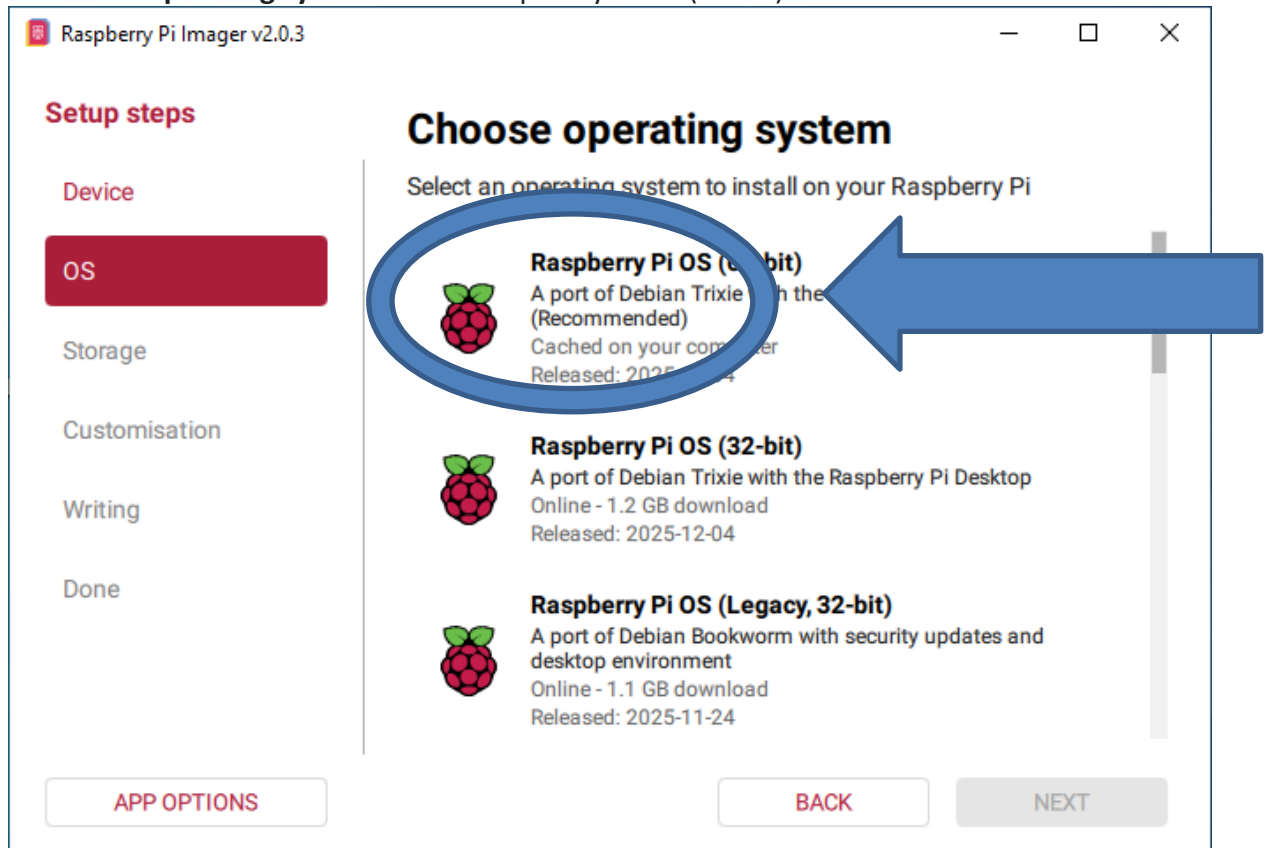
2. Run Raspberry Pi Imager.
3. These instructions are written for **Raspberry Pi Imager v2.0.3** and **Debian Trixie**
3. **Select your Raspberry Pi device** – here an RPi4



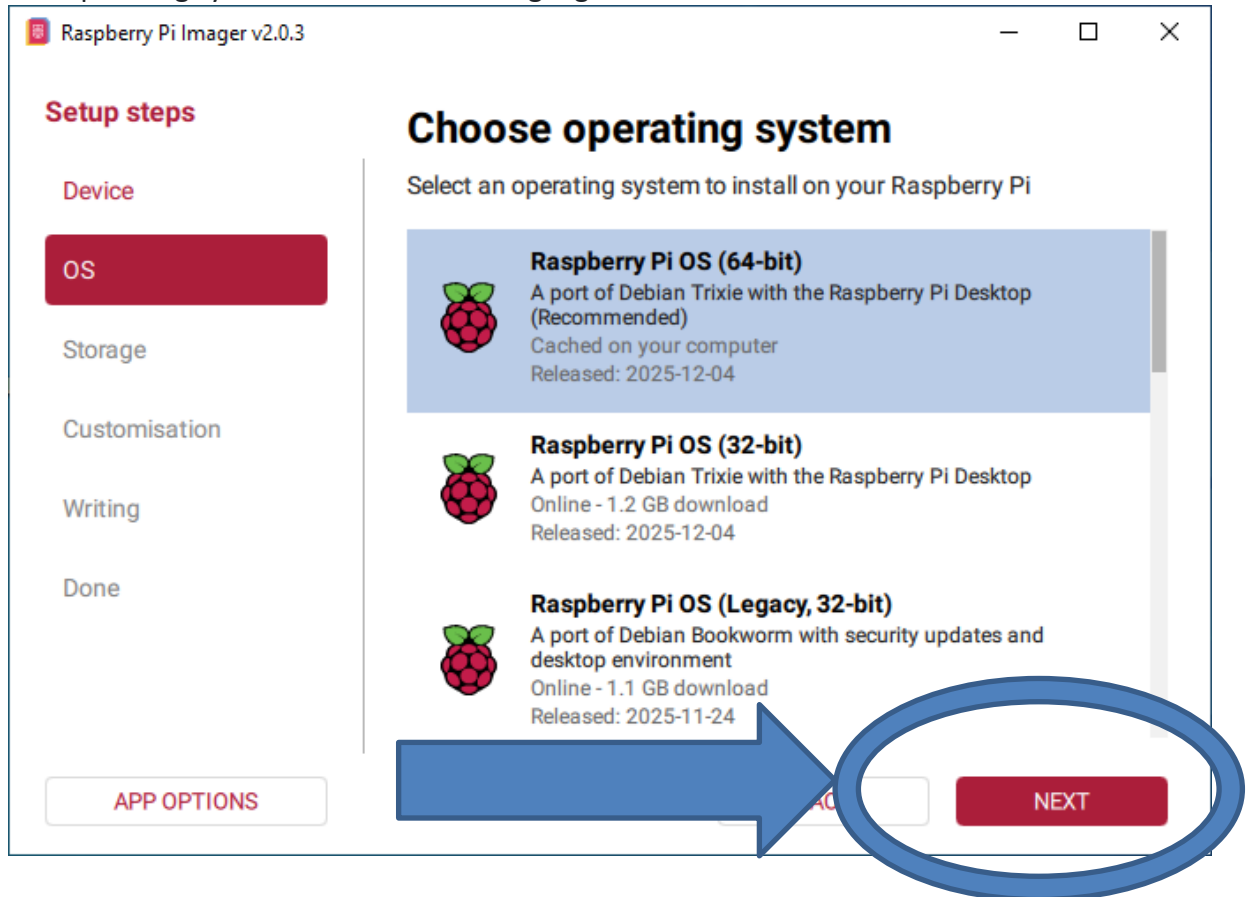
4. The device selected will be highlighter. Then select **NEXT**.



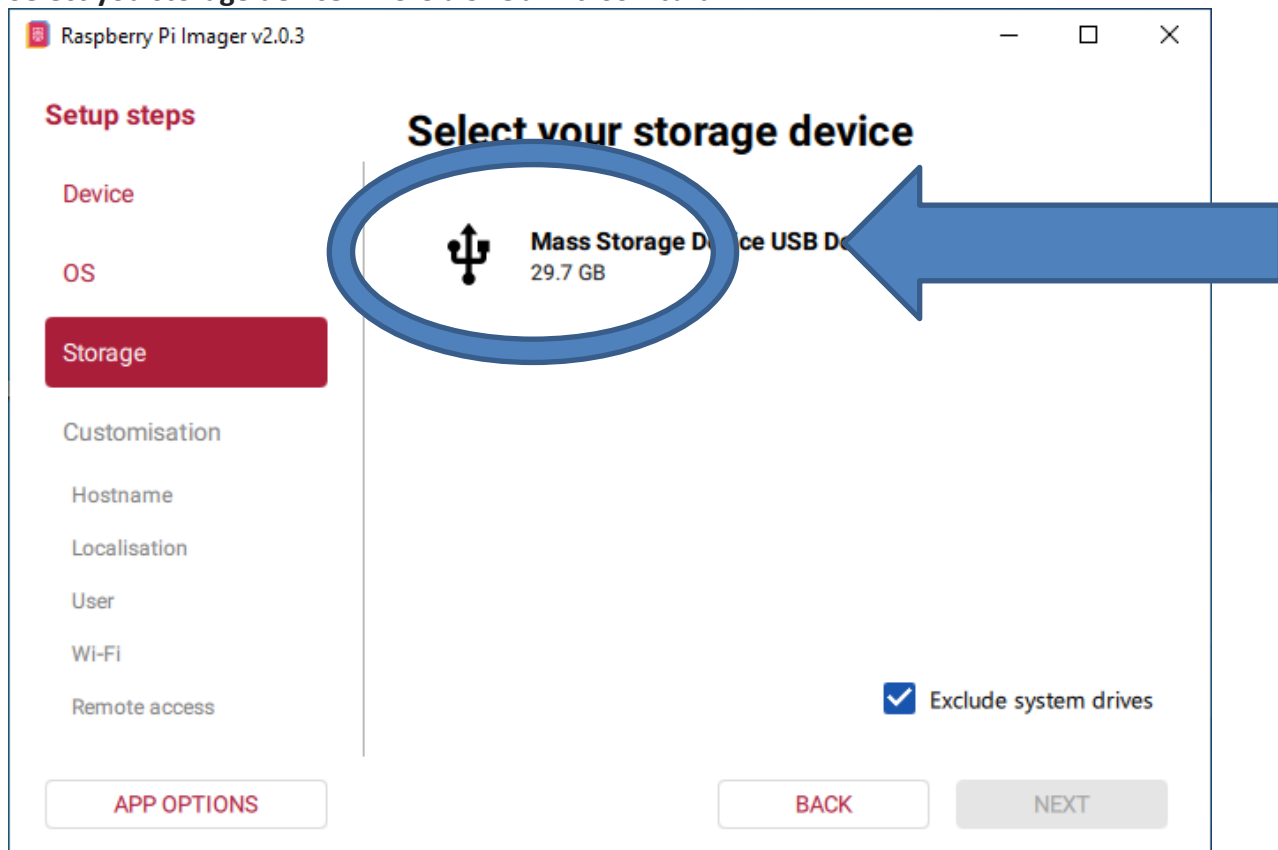
5. **Select the operating system** – here Raspberry Pi OS (64-bit).



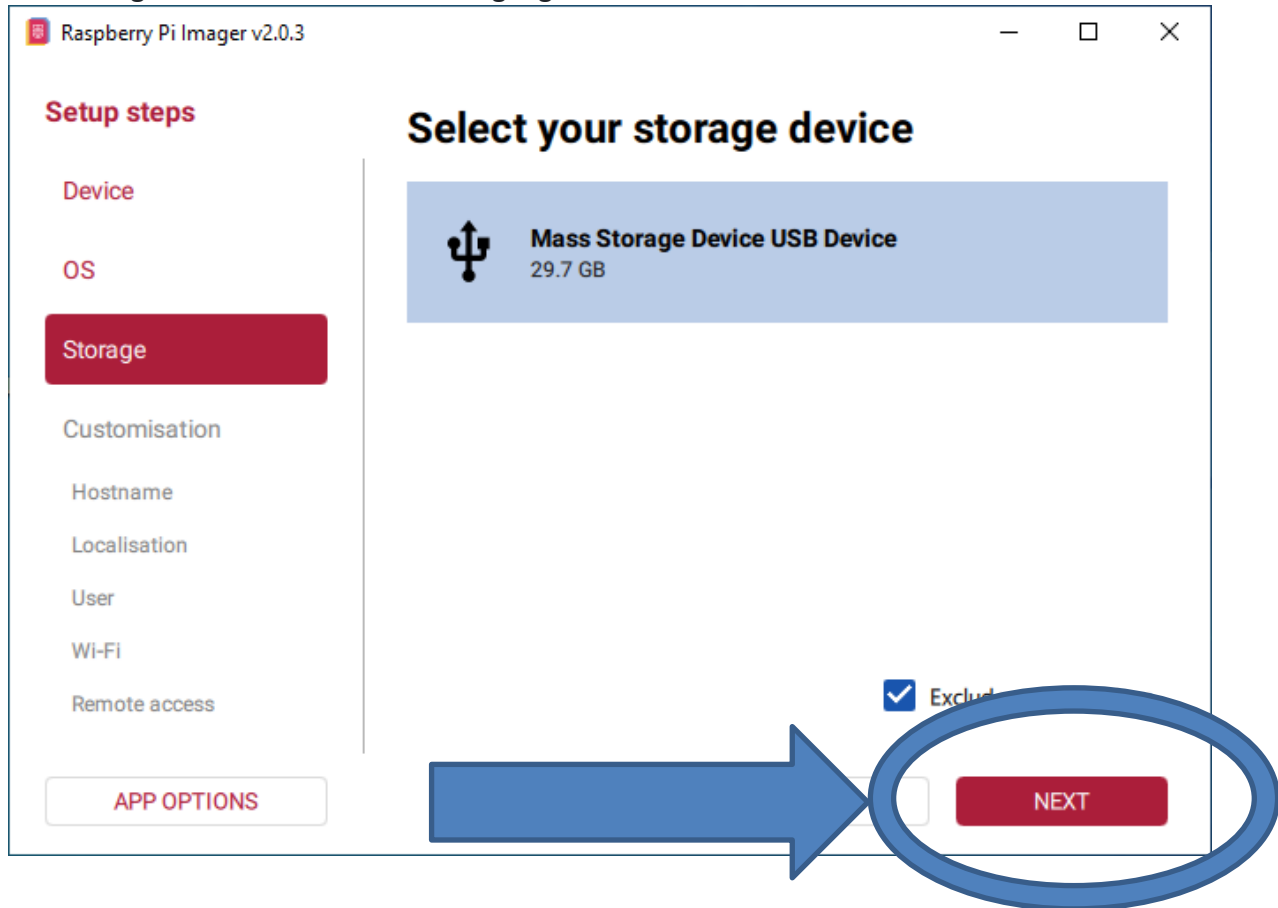
6. The operating system selected will be highlighted – then select **NEXT**.



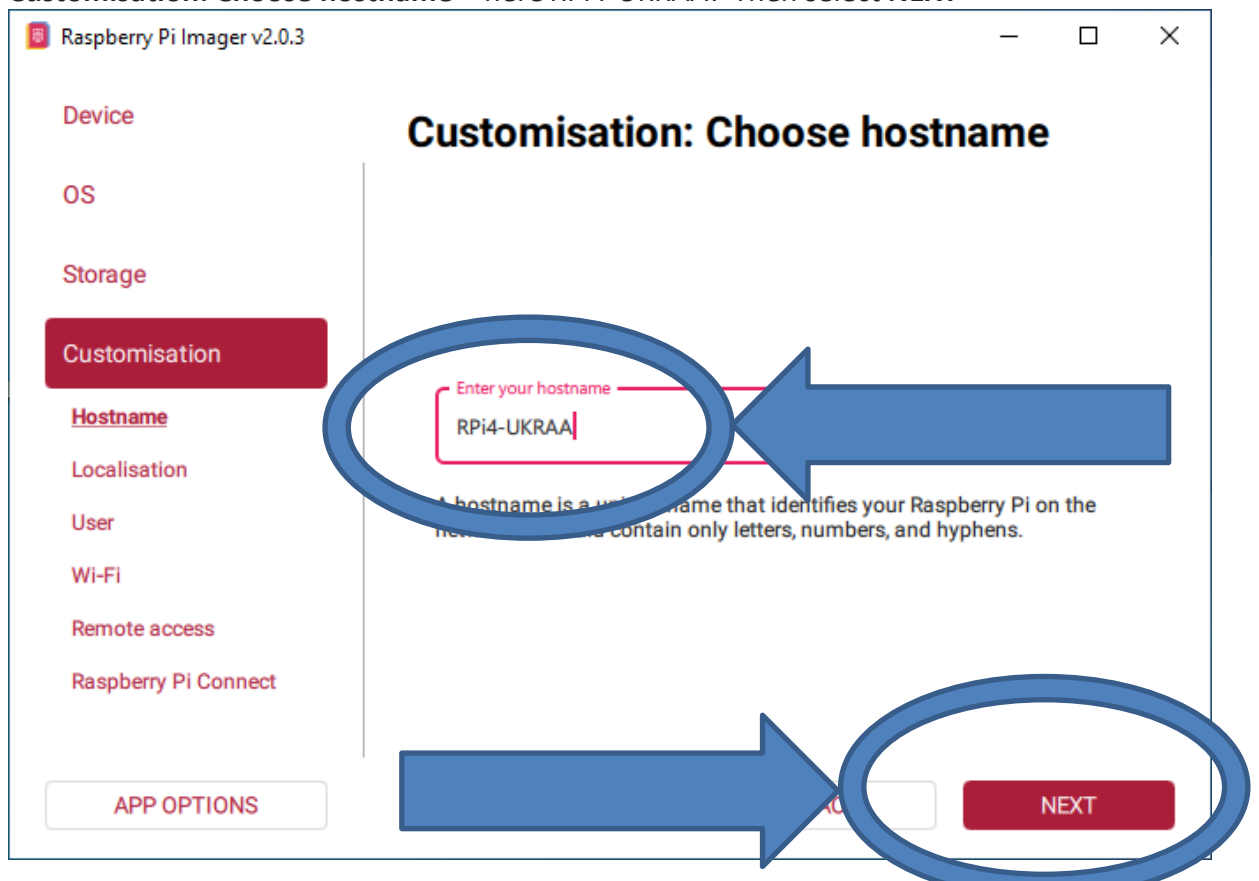
7. Select your storage device – here a 32Gb microSD card.



8. The storage device selected will be highlighted – then select **NEXT**.



9. **Customisation: Choose hostname** – here *RPi4-UKRAA*. Then select **NEXT**



10. **Customisation: Localisation** – change for your specific location. Then select **NEXT**

Raspberry Pi Imager v2.0.3

Customisation: Localisation

Select your location for suggested time zone and keyboard layout

Capital city: London (United Kingdom)

Time zone: Europe/London

Keyboard layout: gb

APP OPTIONS

NEXT

11. **Customisation: Choose username** – leave as **pi**, create password. Then select **NEXT**

Raspberry Pi Imager v2.0.3

Customisation: Choose username

Create a user account for your Raspberry Pi

Enter your username: pi

Password: Saved (hidden) — leave blank to keep

Confirm password: Re-enter to change password

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

APP OPTIONS

NEXT

12. **Customisation: Choose Wi-Fi** – enter your Wi-Fi details. Then select **NEXT**.

Raspberry Pi Imager v2.0.3

Customisation: Choose Wi-Fi

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

SECURE NETWORK **OPEN NETWORK**

SSID: Network name Your_wireless_name

Password: Network password Your_wireless_password

Confirm password: Re-enter password

☐ Hidden SSID

NEXT

13. **Customisation: SSH authentication** – enable SSH. Then select **NEXT**.

Raspberry Pi Imager v2.0.3

Customisation: SSH authentication

Configure SSH access

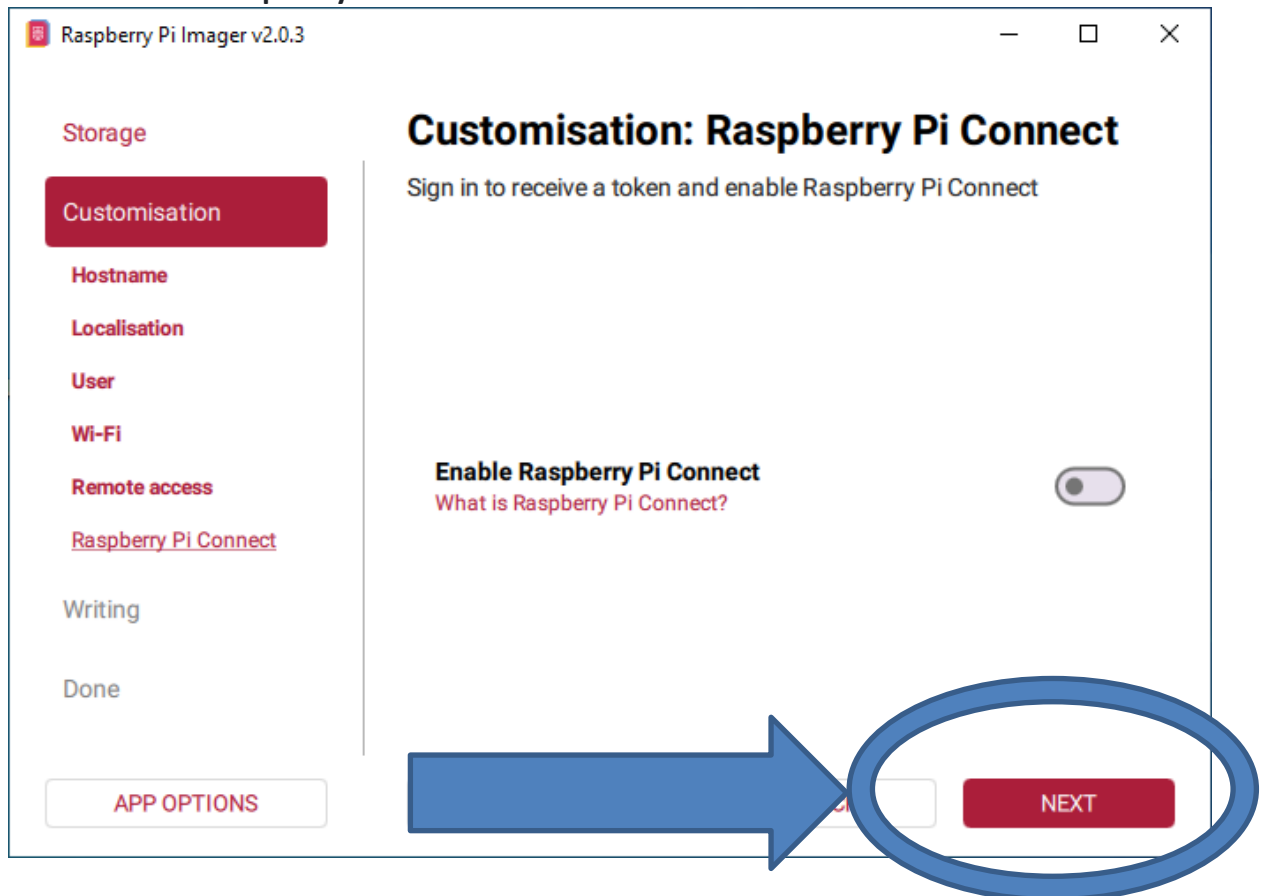
Authentication mechanism:

☒ Use password authentication

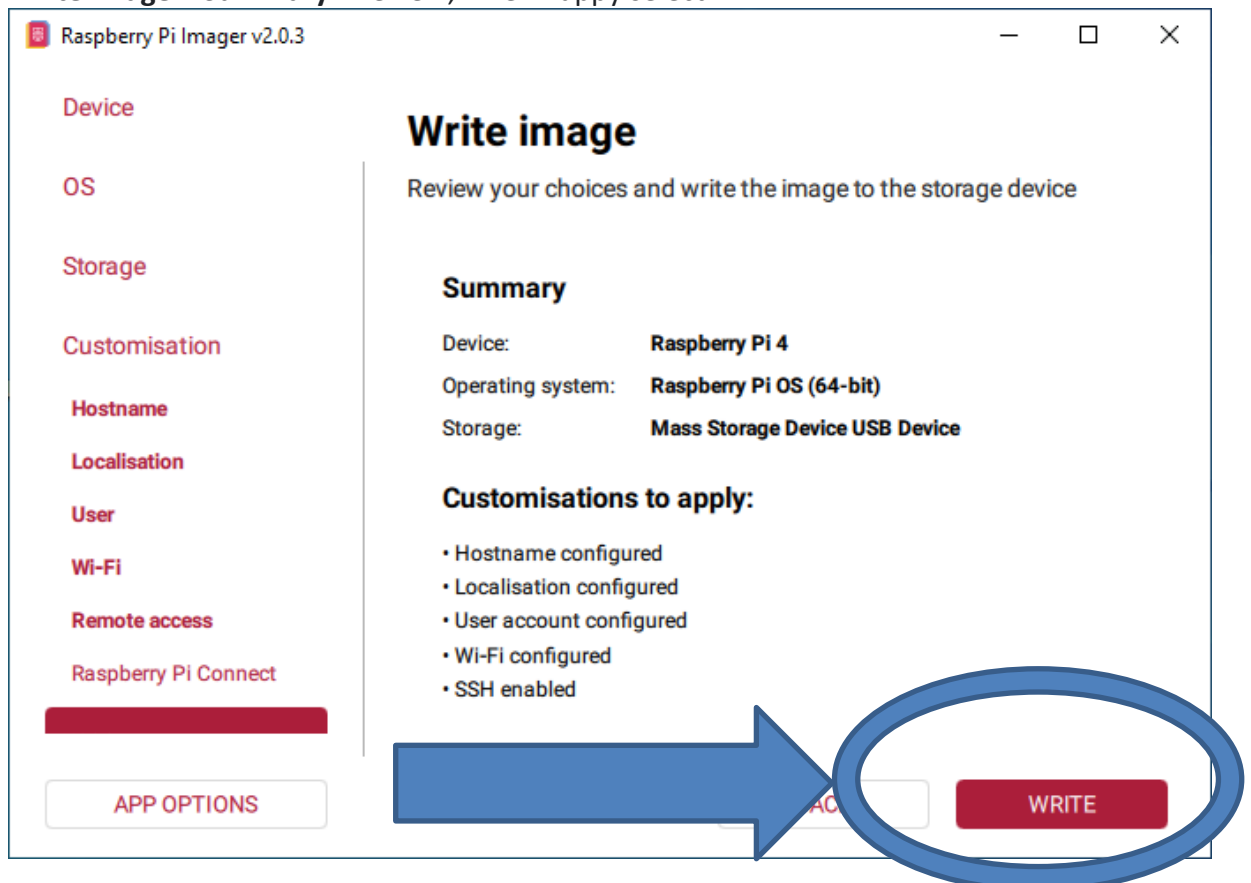
☐ Use public key authentication

NEXT

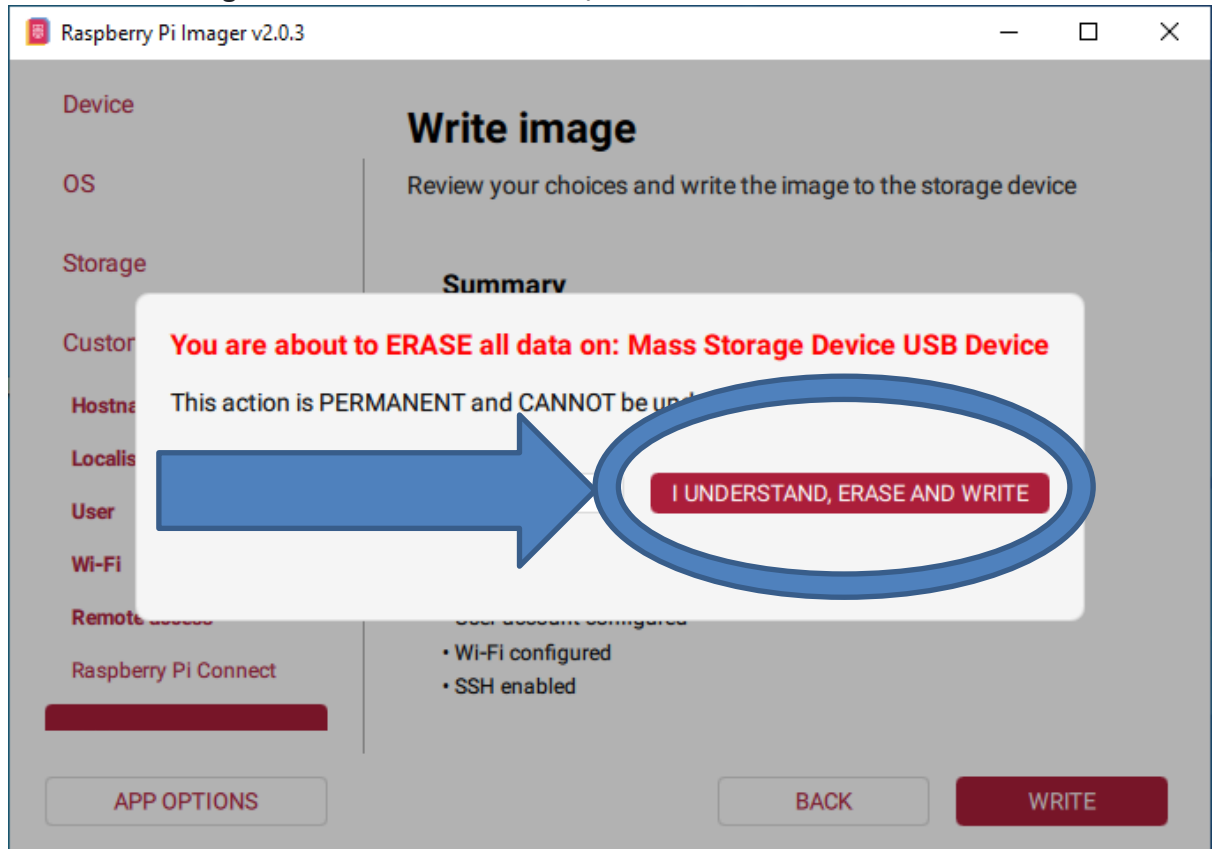
14. **Customisation: Raspberry Pi Connect** – enable if wanted. Then select **NEXT**.



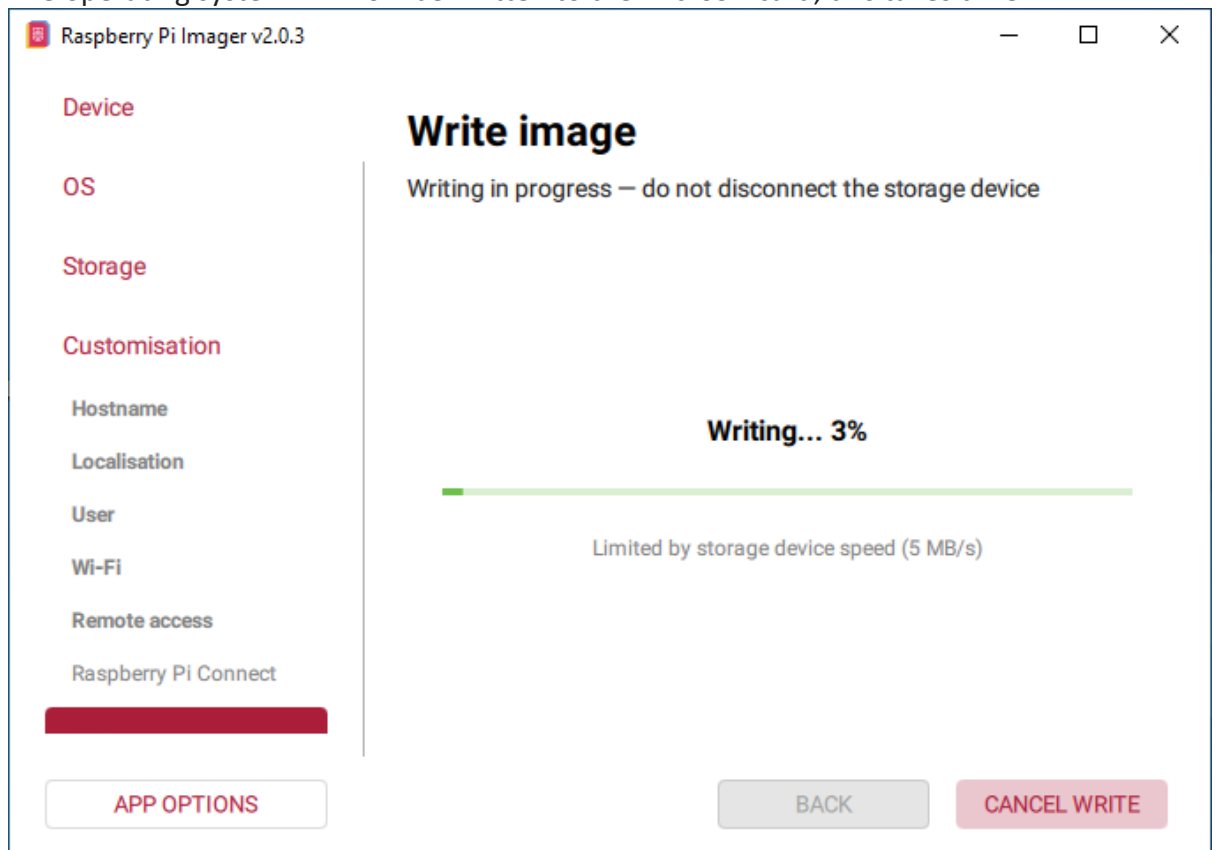
15. **Write image – Summary** – review, when happy select **WRITE**.



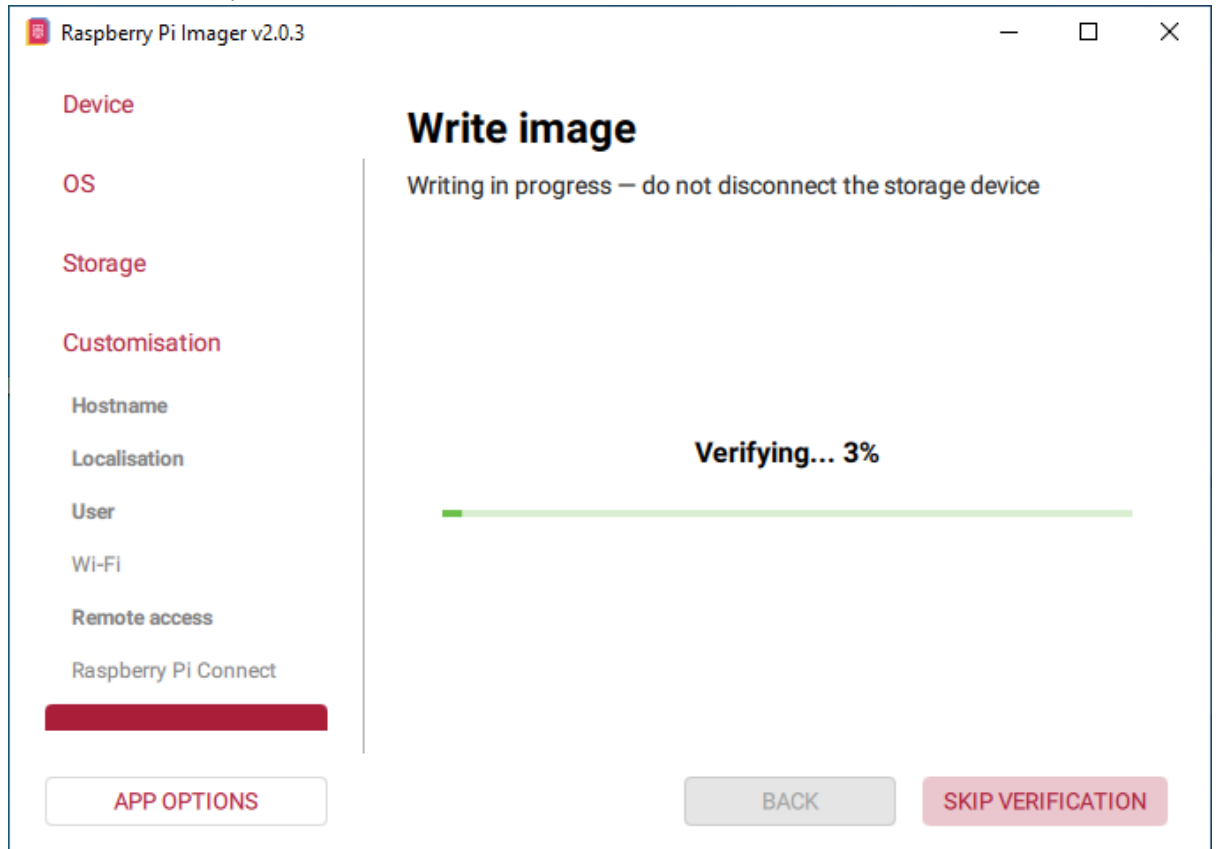
16. Read the warning and select **I UNDERSTAND, ERASE AND WRITE**.



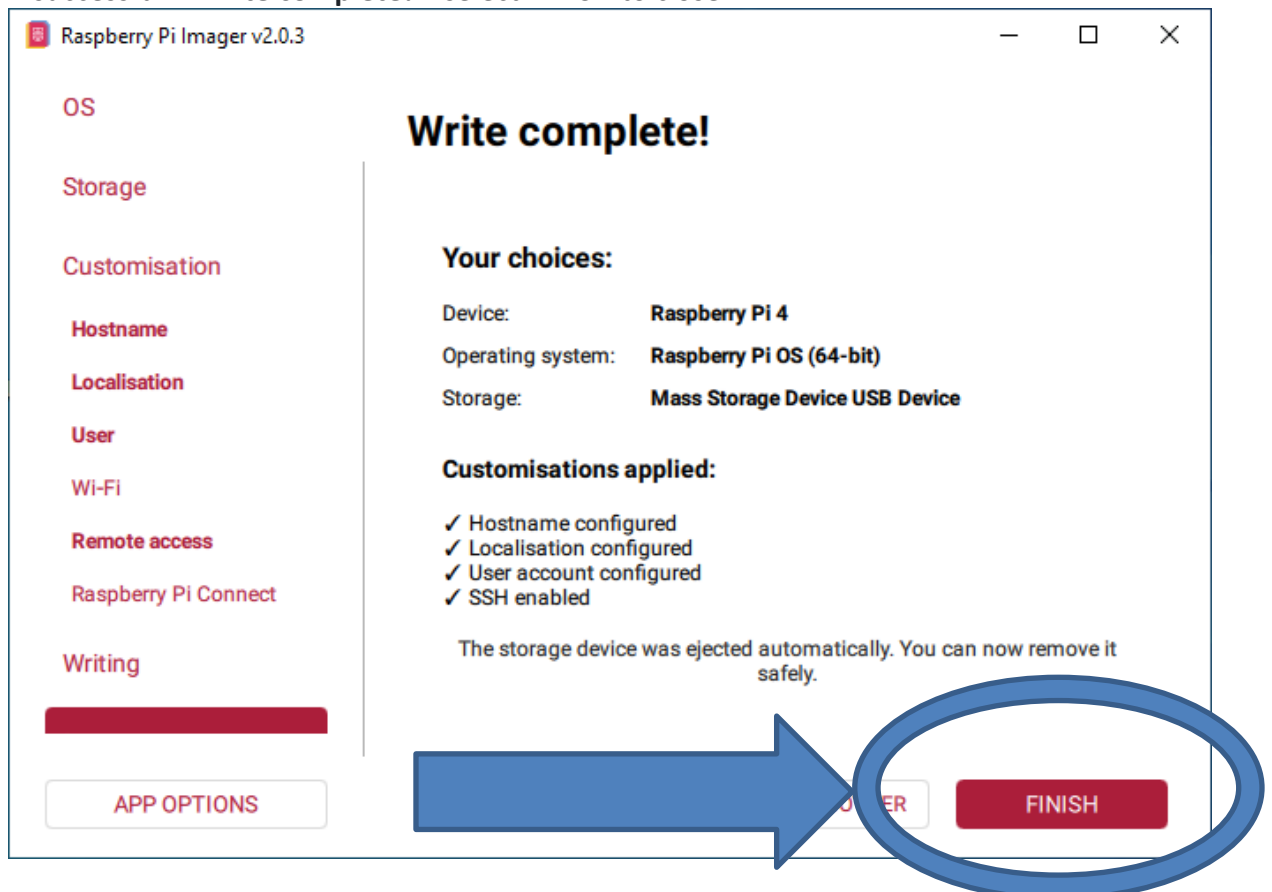
17. The operating system will now be written to the microSD card, this takes time...



18. and then verified, this take a bit more time...



19. If successful – **Write complete!** – select FINISH to close.

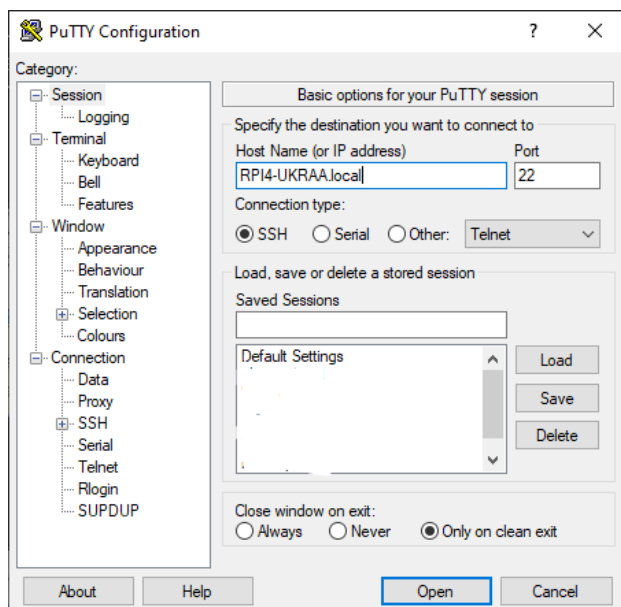


RPi4 first boot and initial configuration

Put the microSD card you have just created into your RPi4, connect the RPi to your LAN if necessary, insert the power cable and turn on.

Using PuTTY (<https://www.putty.org/>)

1. In **Host Name (or IP address)**, type the host name of your RPi that you wrote down earlier... and select **Open**

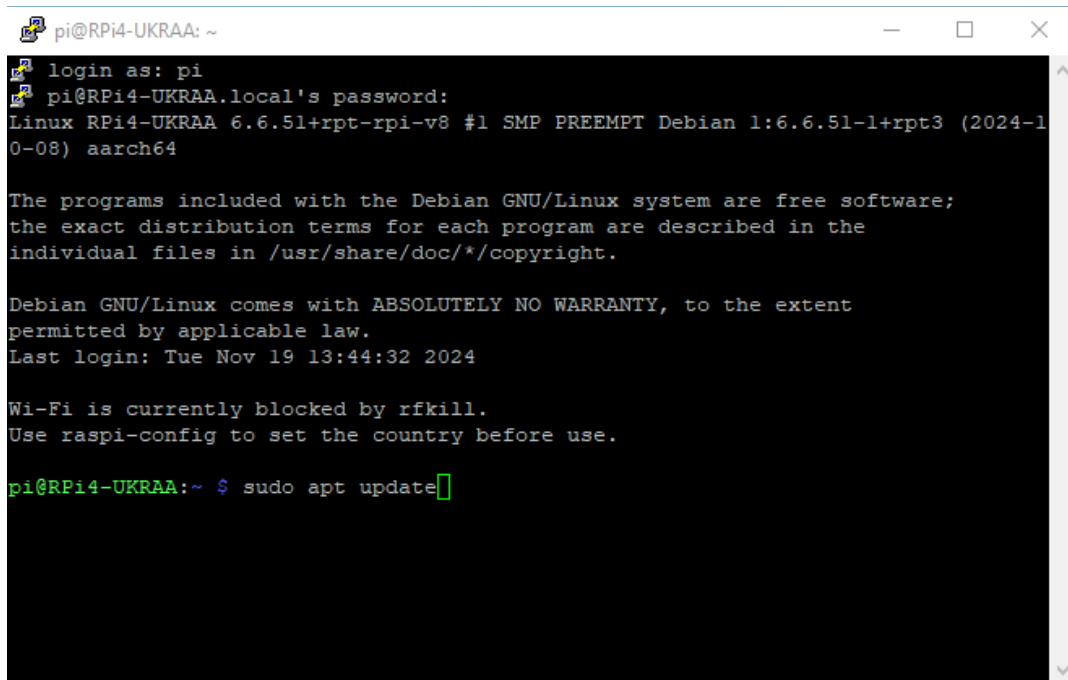


2. You will be presented with a PuTTY Security Alert - select **Accept**. You will now have a PuTTY terminal window.



3. Login as **pi** and enter your **password**
4. Now type the command below and press **enter**. This will update the RPi OS

sudo apt update

A terminal window titled 'pi@RPi4-UKRAA: ~' showing the login process for the 'pi' user. The prompt is 'pi@RPi4-UKRAA.local's password:'. The system output includes the Linux version 'Linux RPi4-UKRAA 6.6.51+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.6.51-1+rpt3 (2024-10-08) aarch64', a disclaimer about Debian GNU/Linux being free software, and the last login time 'Tue Nov 19 13:44:32 2024'. The user then enters the command 'sudo apt update' at the prompt 'pi@RPi4-UKRAA:~ \$'.

```
pi@RPi4-UKRAA: ~
login as: pi
pi@RPi4-UKRAA.local's password:
Linux RPi4-UKRAA 6.6.51+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.6.51-1+rpt3 (2024-10-08) aarch64

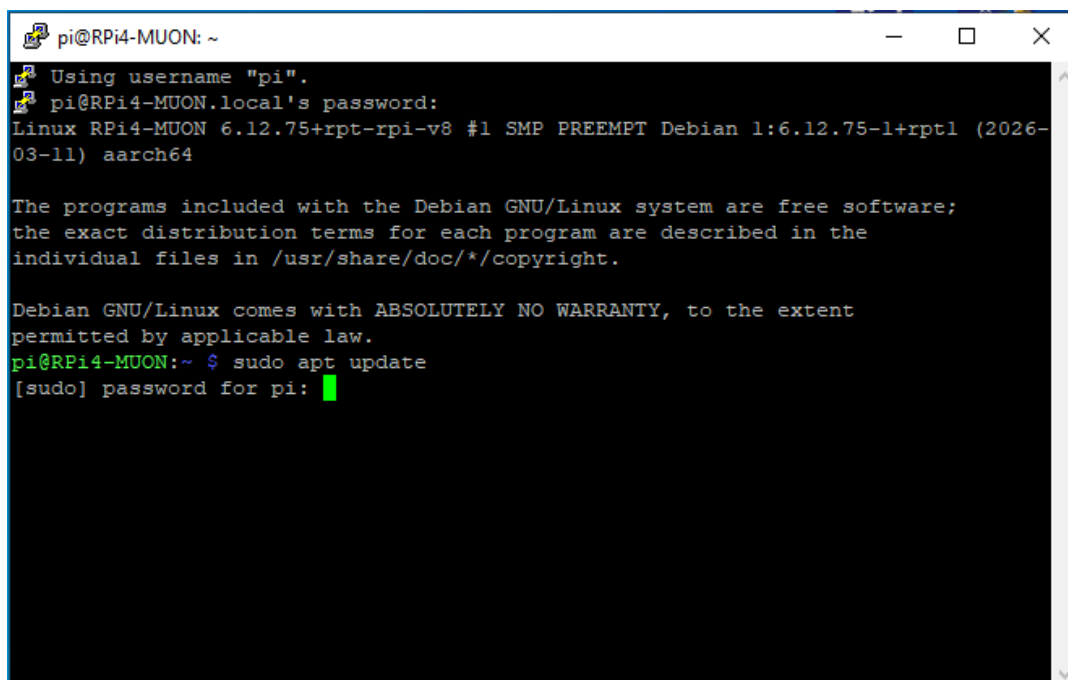
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Nov 19 13:44:32 2024

Wi-Fi is currently blocked by rfkill.
Use raspi-config to set the country before use.

pi@RPi4-UKRAA:~ $ sudo apt update
```

5. You will be asked for your password, enter your **password** and press enter.

A terminal window titled 'pi@RPi4-MUON: ~' showing the login process for the 'pi' user. The prompt is 'pi@RPi4-MUON.local's password:'. The system output includes the Linux version 'Linux RPi4-MUON 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64', a disclaimer about Debian GNU/Linux being free software, and the last login time 'Tue Nov 19 13:44:32 2024'. The user then enters the command 'sudo apt update' at the prompt 'pi@RPi4-MUON:~ \$'. The prompt changes to '[sudo] password for pi:' with a redacted password field.

```
pi@RPi4-MUON: ~
Using username "pi".
pi@RPi4-MUON.local's password:
Linux RPi4-MUON 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.

pi@RPi4-MUON:~ $ sudo apt update
[sudo] password for pi: 
```

- Now type the command below and press **enter**. This will update any preinstalled software packages. You will be asked to type **y** and press **enter** to proceed.

sudo apt upgrade

```
pi@RPi4-MUON: ~  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Wed May 6 12:38:04 2026 from 2a00:23cc:c64b:8f01:c0dd:345e:e284:c3d  
4  
pi@RPi4-MUON:~ $ sudo apt update  
[sudo] password for pi:  
Hit:1 http://deb.debian.org/debian trixie InRelease  
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]  
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]  
Get:4 http://archive.raspberrypi.com/debian trixie InRelease [54.9 kB]  
Get:5 http://deb.debian.org/debian-security trixie-security/main arm64 Packages  
[130 kB]  
Get:6 http://archive.raspberrypi.com/debian trixie/main armhf Packages [433 kB]  
Get:7 http://deb.debian.org/debian-security trixie-security/main armhf Packages  
[123 kB]  
Get:8 http://archive.raspberrypi.com/debian trixie/main arm64 Packages [438 kB]  
Get:9 http://deb.debian.org/debian-security trixie-security/main Translation-en  
[82.8 kB]  
Fetched 1,262 kB in 1s (2,172 kB/s)  
13 packages can be upgraded. Run 'apt list --upgradable' to see them.  
pi@RPi4-MUON:~ $ sudo apt upgrade
```

- You may be advised that you can remove **no longer required** packages. You can type the command below and press **enter**. Again, you will be asked to type **y** and press **enter** to proceed.

sudo apt autoremove

```
pi@RPi4-MUON: ~  
Setting up libpackagekit-glib2-18:arm64 (1.3.1-1+deb13u1) ...  
Setting up rpinters (1.8) ...  
Setting up python3-sense-hat (2.6.1-1) ...  
Setting up ntfs-3g (1:2022.10.3-5+deb13u1) ...  
Setting up libmagickcore-7.q16-10:arm64 (8:7.1.1.43+dfsg1-1+deb13u8) ...  
Setting up packagekit (1.3.1-1+deb13u1) ...  
Setting up libngtcp2-16:arm64 (1.11.0-1+deb13u1) ...  
Setting up libngtcp2-crypto-gnutls8:arm64 (1.11.0-1+deb13u1) ...  
Setting up libmagickwand-7.q16-10:arm64 (8:7.1.1.43+dfsg1-1+deb13u8) ...  
Setting up libmagickcore-7.q16-10-extra:arm64 (8:7.1.1.43+dfsg1-1+deb13u8) ...  
Processing triggers for initramfs-tools (0.148.3+rpt2) ...  
update-initramfs: Generating /boot/initrd.img-6.12.75+rpt-rpi-v8  
'/boot/initrd.img-6.12.75+rpt-rpi-v8' -> '/boot/firmware/initramfs8'  
update-initramfs: Generating /boot/initrd.img-6.12.75+rpt-rpi-2712  
'/boot/initrd.img-6.12.75+rpt-rpi-2712' -> '/boot/firmware/initramfs_2712'  
Processing triggers for hicolor-icon-theme (0.18-2) ...  
Processing triggers for libc-bin (2.41-12+rpt1+deb13u2) ...  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for dbus (1.16.2-2) ...  
Processing triggers for mailcap (3.74) ...  
Processing triggers for desktop-file-utils (0.28-1) ...  
pi@RPi4-MUON:~ $ sudo apt autoremove
```


8. We can now set up VNC to access the RPi from our desktop PC. Type the command below and press **enter**. We will be presented with the RPi configuration tool.

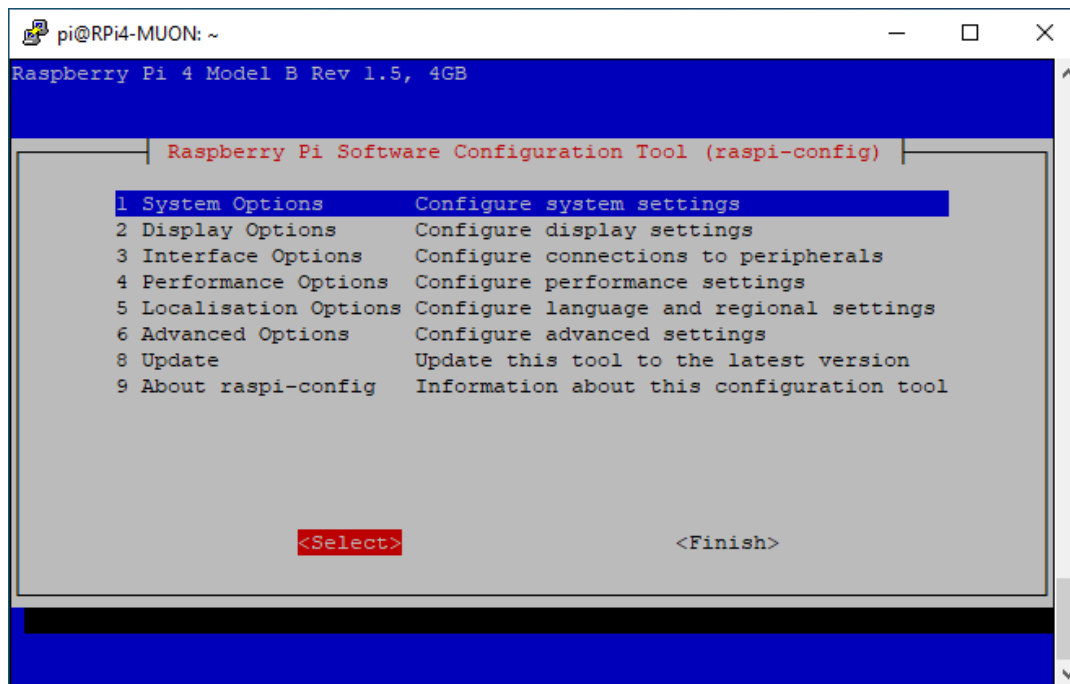
sudo raspi-config

```
pi@RPi4-MUON: ~  
Setting up python3-sense-hat (2.6.1-1) ...  
Setting up ntfs-3g (1:2022.10.3-5+deb13u1) ...  
Setting up libmagickcore-7.q16-10:arm64 (8:7.1.1.43+dfsg1-1+deb13u8) ...  
Setting up packagekit (1.3.1-1+deb13u1) ...  
Setting up libngtcp2-l6:arm64 (1.11.0-1+deb13u1) ...  
Setting up libngtcp2-crypto-gnutls8:arm64 (1.11.0-1+deb13u1) ...  
Setting up libmagickwand-7.q16-10:arm64 (8:7.1.1.43+dfsg1-1+deb13u8) ...  
Setting up libmagickcore-7.q16-10-extra:arm64 (8:7.1.1.43+dfsg1-1+deb13u8) ...  
Processing triggers for initramfs-tools (0.148.3+rpt2) ...  
update-initramfs: Generating /boot/initrd.img-6.12.75+rpt-rpi-v8  
'/boot/initrd.img-6.12.75+rpt-rpi-v8' -> '/boot/firmware/initramfs8'  
update-initramfs: Generating /boot/initrd.img-6.12.75+rpt-rpi-2712  
'/boot/initrd.img-6.12.75+rpt-rpi-2712' -> '/boot/firmware/initramfs_2712'  
Processing triggers for hicolor-icon-theme (0.18-2) ...  
Processing triggers for libc-bin (2.41-12+rpt1+deb13u2) ...  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for dbus (1.16.2-2) ...  
Processing triggers for mailcap (3.74) ...  
Processing triggers for desktop-file-utils (0.28-1) ...  
pi@RPi4-MUON:~ $ sudo apt autoremove  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
pi@RPi4-MUON:~ $ sudo raspi-config  
pi@RPi4-MUON:~ $ sudo raspi-config
```

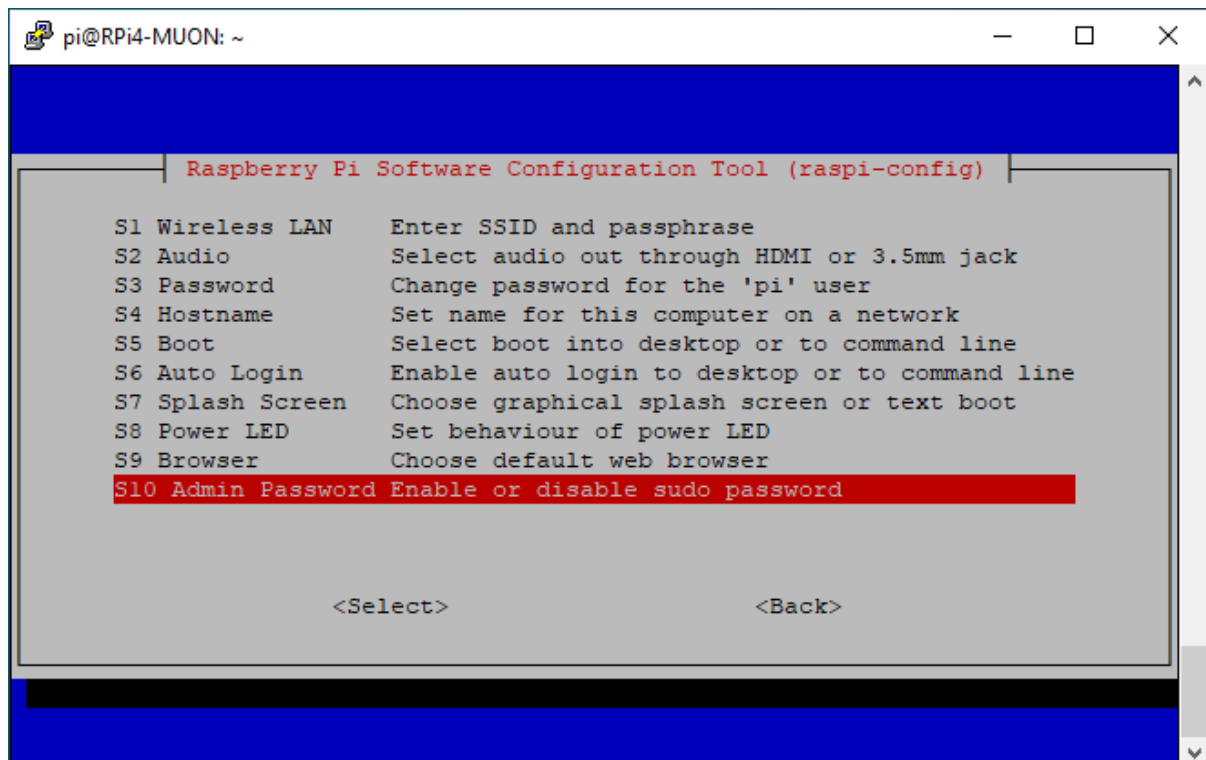
9. Select 1 System Options

```
pi@RPi4-MUON: ~  
Raspberry Pi 4 Model B Rev 1.5, 4GB  
  
Raspberry Pi Software Configuration Tool (raspi-config)  
  
1 System Options      Configure system settings  
2 Display Options     Configure display settings  
3 Interface Options   Configure connections to peripherals  
4 Performance Options Configure performance settings  
5 Localisation Options Configure language and regional settings  
6 Advanced Options    Configure advanced settings  
8 Update              Update this tool to the latest version  
9 About raspi-config  Information about this configuration tool  
  
<Select>              <Finish>
```

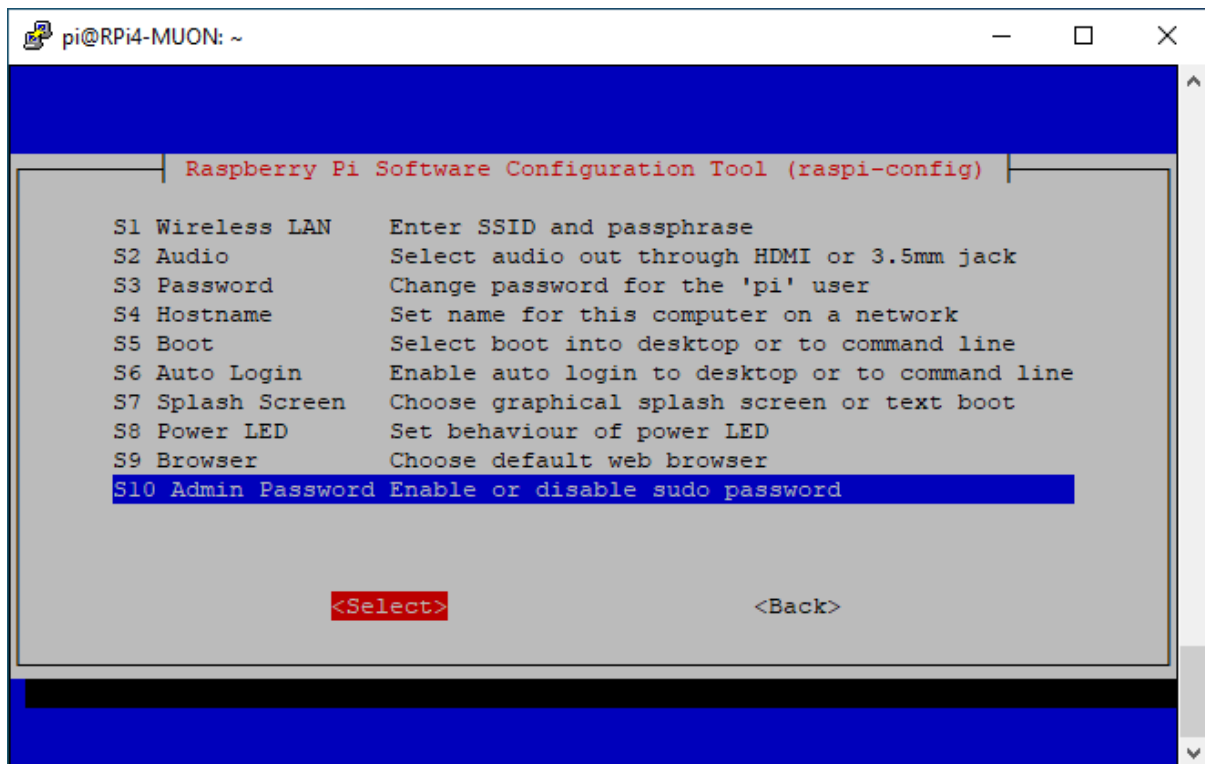
10. Navigate to **Select** and press **enter**.



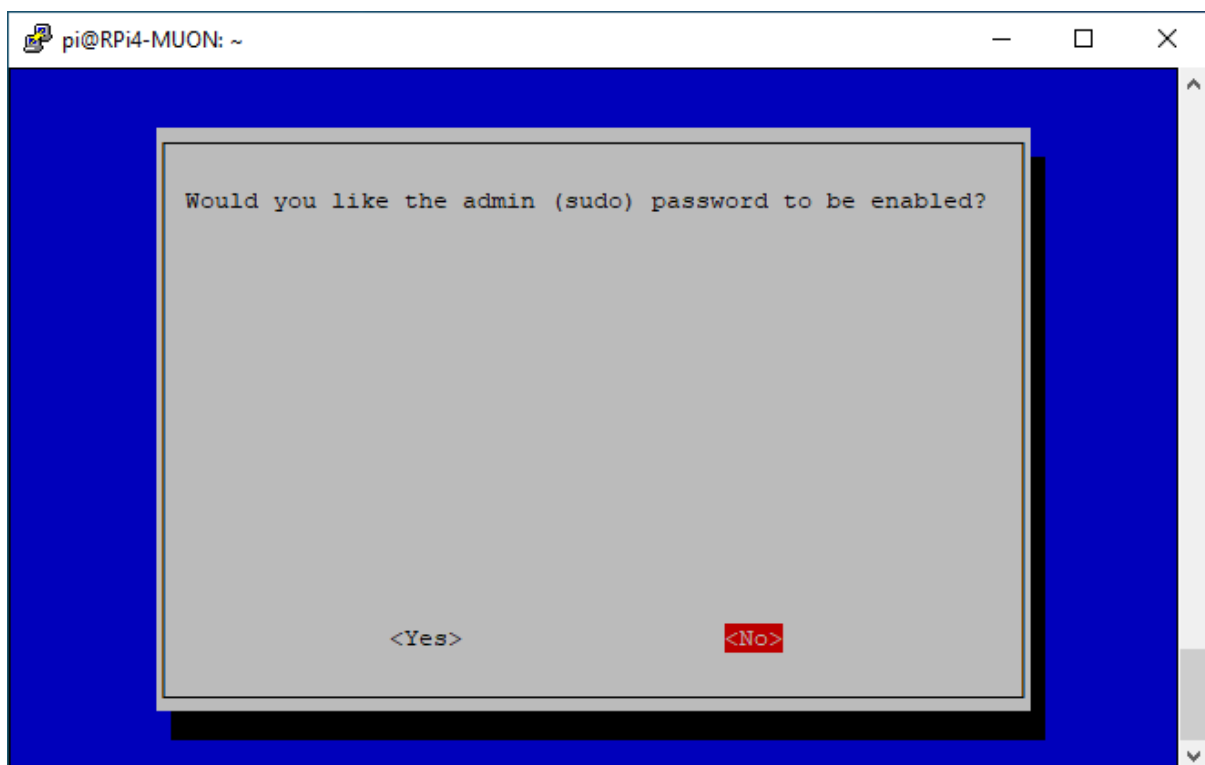
11. Select **S10 Admin Password**



12. Navigate to **Select** and press **enter**.



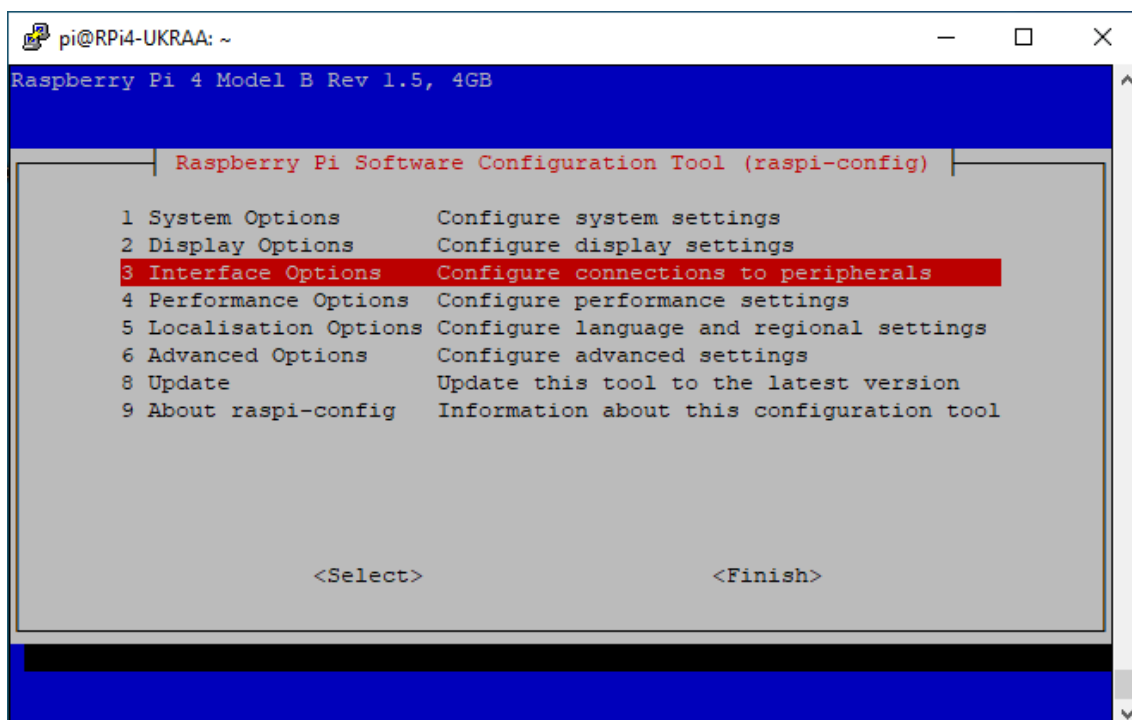
13. Select **No** and press **enter** to disable admin (sudo) password.



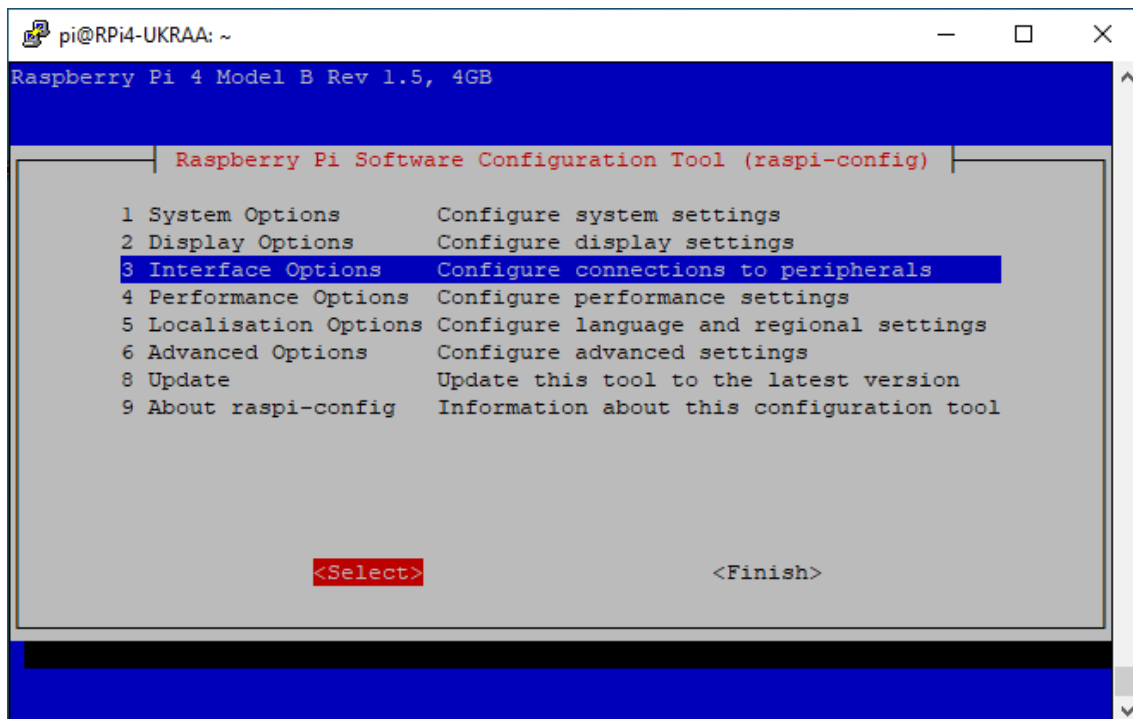
14. You will be informed that the admin password is disabled, press **enter**.



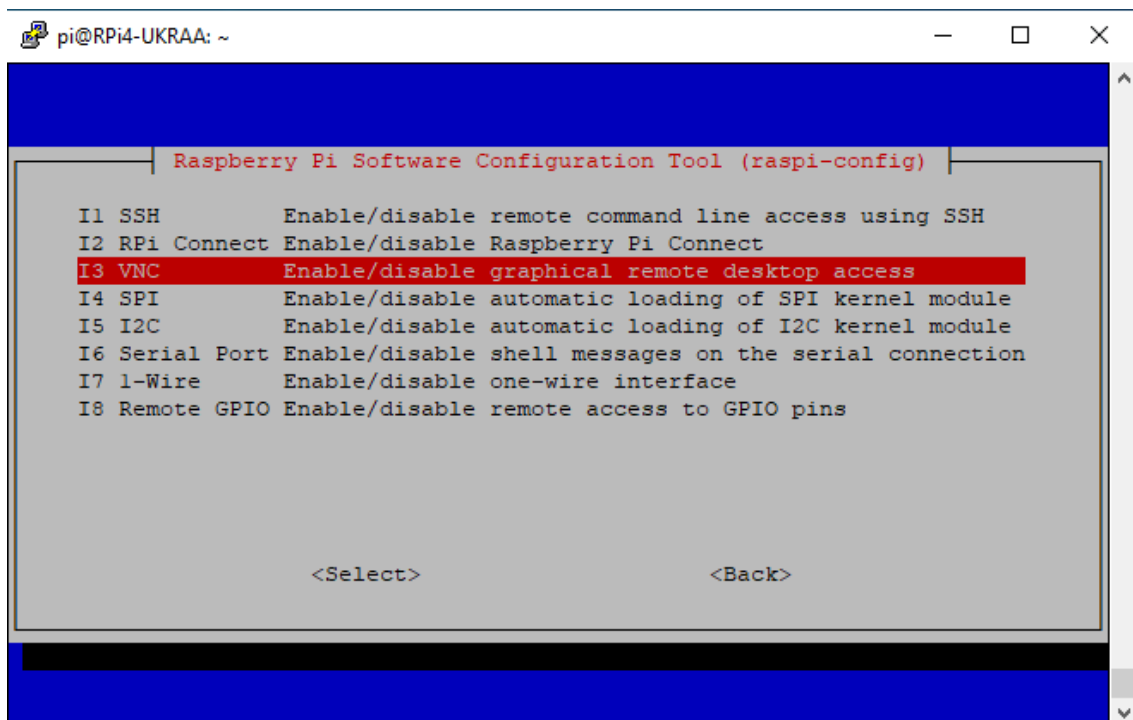
15. Select **I3 Interface Options**



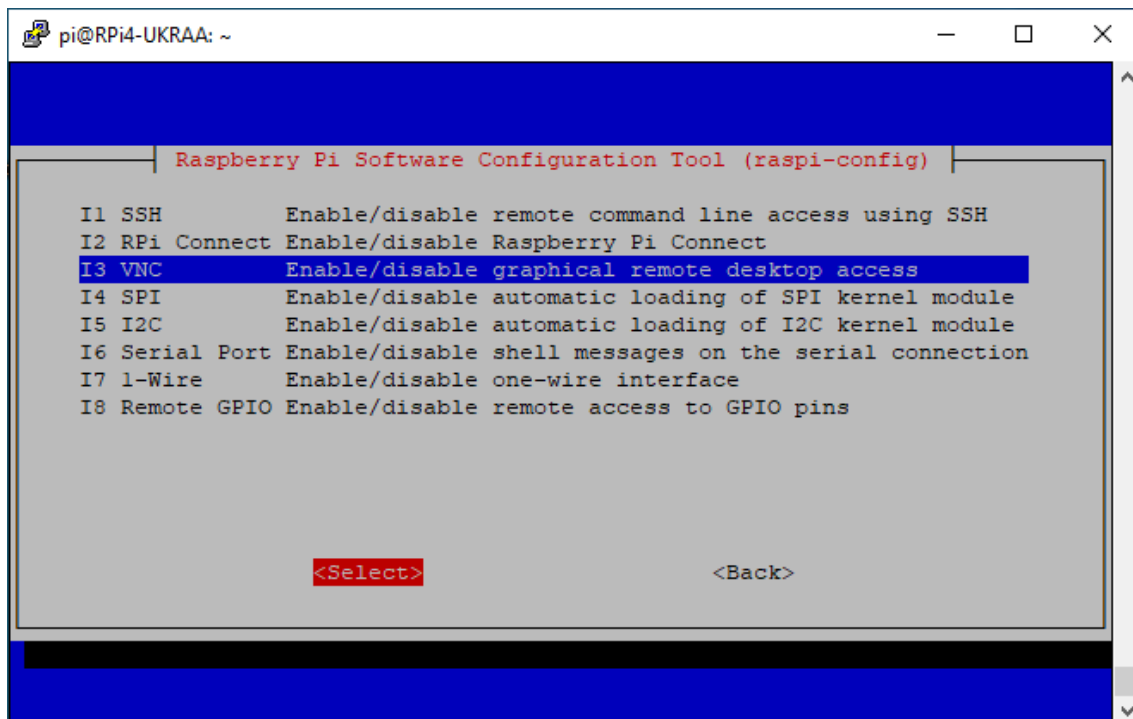
16. Navigate to **Select** and press **enter**.



17. Now navigate down to **I3 VNC**.



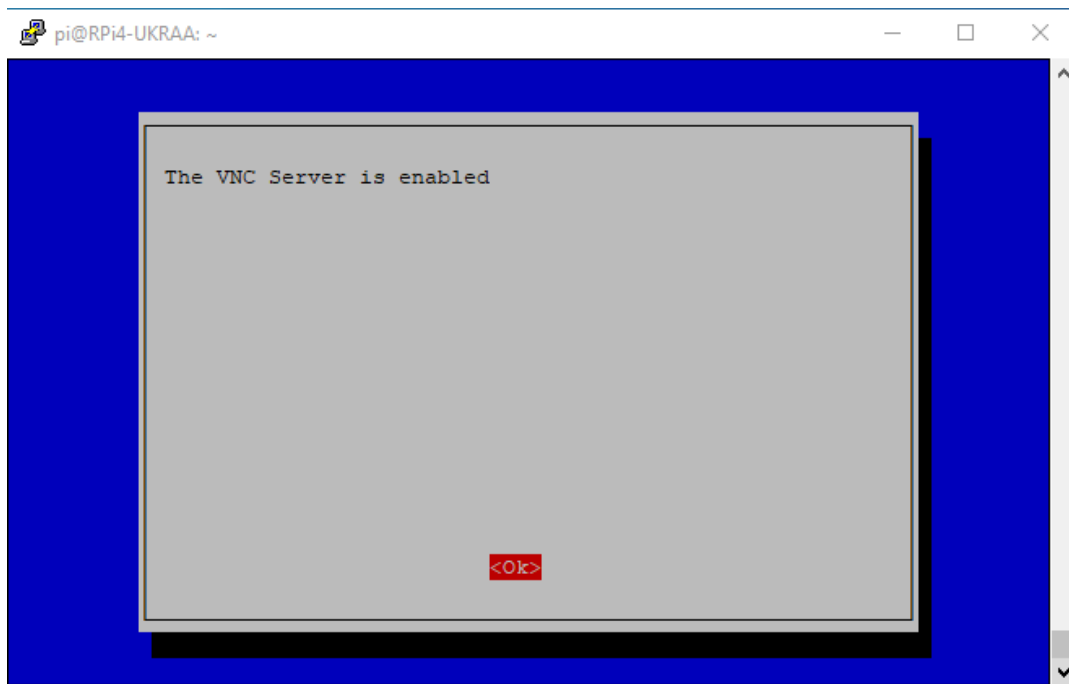
18. Navigate to **Select** and press **enter**.



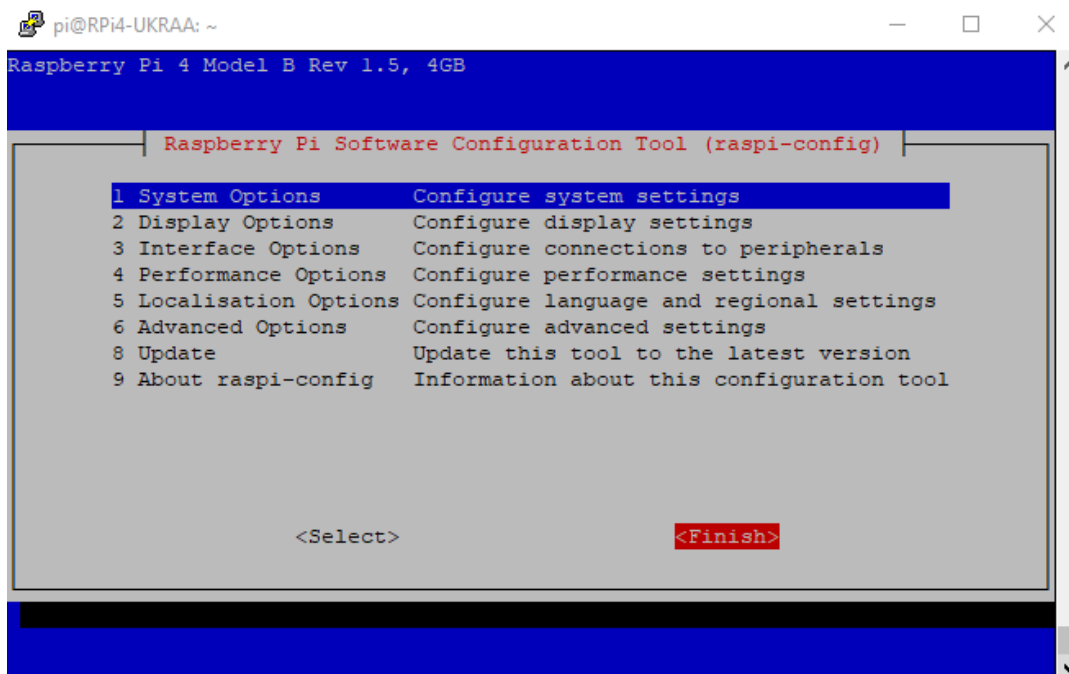
19. Select **Yes** and press **enter** to enable the VNC Server.



20. VNC server will now be enabled, press **enter**.

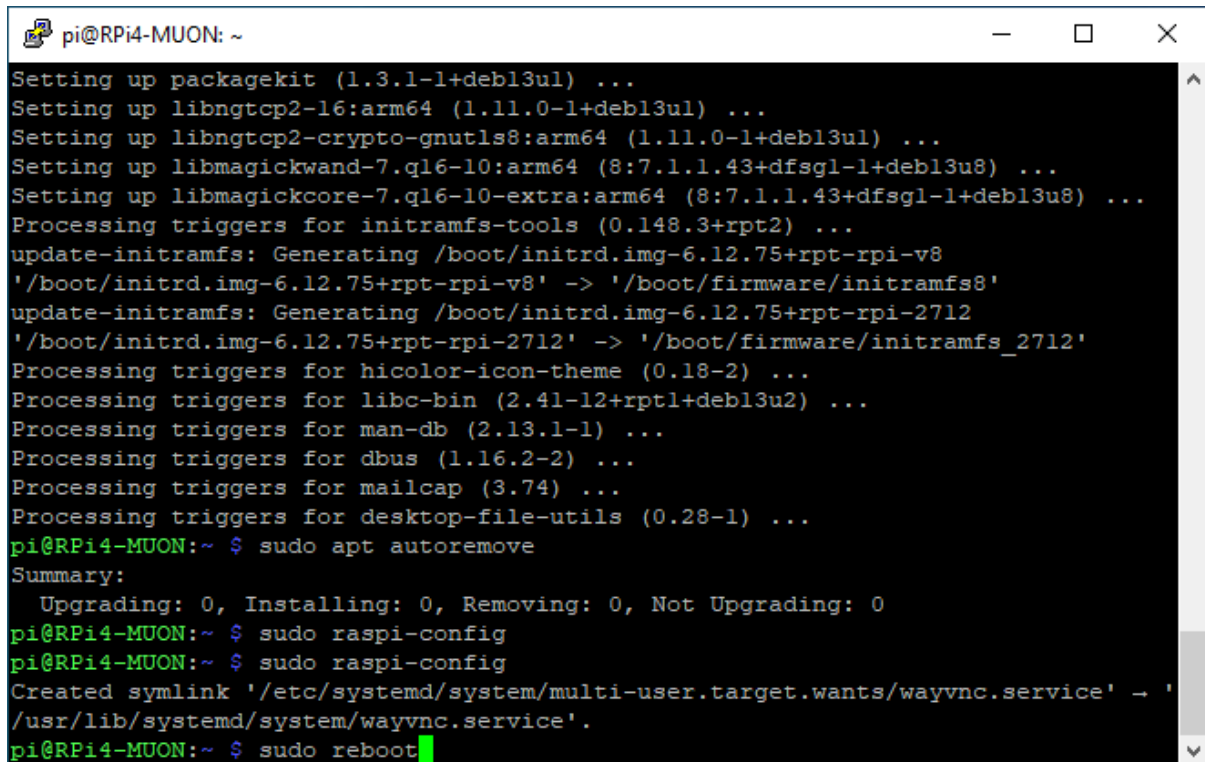


21. Finish with the Configuration Tool by selecting **Finish** and press **enter**.



22. Reboot the RPi, type the command below and press **enter**. This will close the PuTTY connection. You can close the Putty window.

sudo reboot

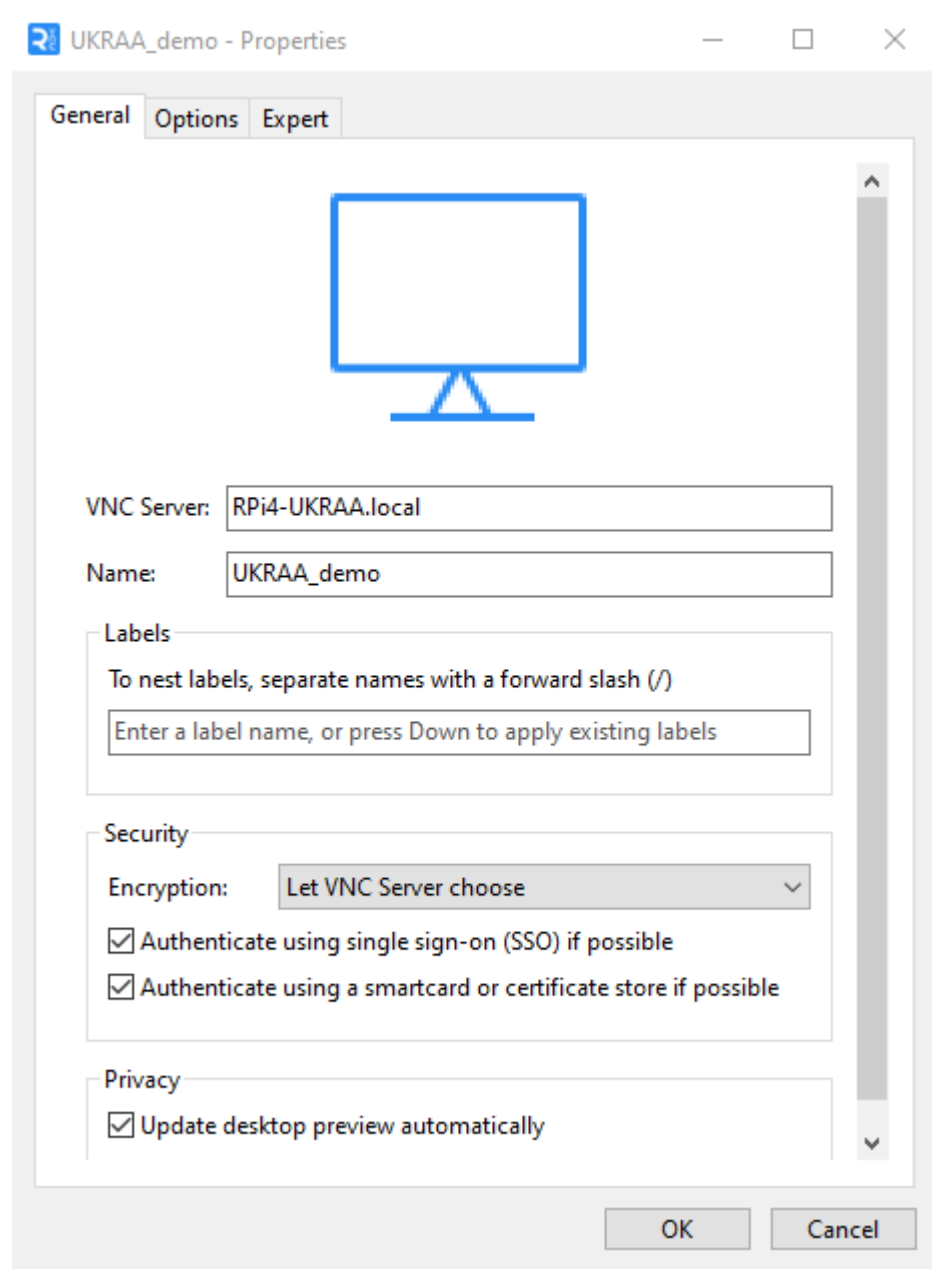


```
pi@RPi4-MUON: ~  
Setting up packagekit (1.3.1-1+deb13u1) ...  
Setting up libngtcp2-16:arm64 (1.11.0-1+deb13u1) ...  
Setting up libngtcp2-crypto-gnutls8:arm64 (1.11.0-1+deb13u1) ...  
Setting up libmagickwand-7.q16-10:arm64 (8:7.1.1.43+dfsg1-1+deb13u8) ...  
Setting up libmagickcore-7.q16-10-extra:arm64 (8:7.1.1.43+dfsg1-1+deb13u8) ...  
Processing triggers for initramfs-tools (0.148.3+rpt2) ...  
update-initramfs: Generating /boot/initrd.img-6.12.75+rpt-rpi-v8  
'/boot/initrd.img-6.12.75+rpt-rpi-v8' -> '/boot/firmware/initramfs8'  
update-initramfs: Generating /boot/initrd.img-6.12.75+rpt-rpi-2712  
'/boot/initrd.img-6.12.75+rpt-rpi-2712' -> '/boot/firmware/initramfs_2712'  
Processing triggers for hicolor-icon-theme (0.18-2) ...  
Processing triggers for libc-bin (2.41-12+rpt1+deb13u2) ...  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for dbus (1.16.2-2) ...  
Processing triggers for mailcap (3.74) ...  
Processing triggers for desktop-file-utils (0.28-1) ...  
pi@RPi4-MUON:~ $ sudo apt autoremove  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
pi@RPi4-MUON:~ $ sudo raspi-config  
pi@RPi4-MUON:~ $ sudo raspi-config  
Created symlink '/etc/systemd/system/multi-user.target.wants/wayvnc.service' -> '/usr/lib/systemd/system/wayvnc.service'.  
pi@RPi4-MUON:~ $ sudo reboot
```

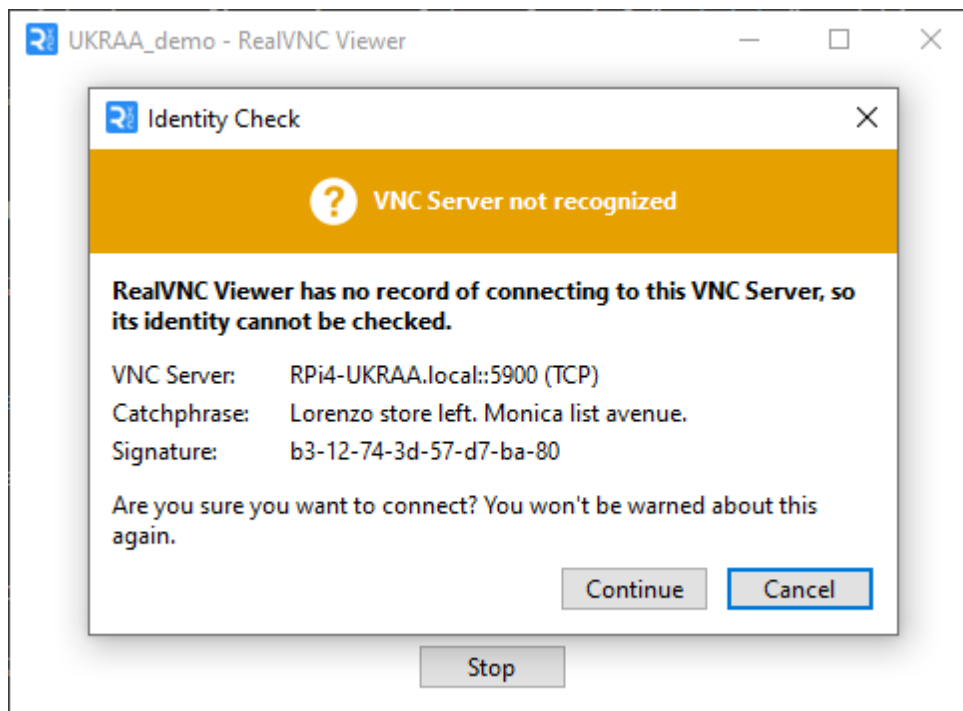

Headless access to RPi via VNC

Using a VNC program like RealVNC (<https://www.realvnc.com/en/>), connect to your RPi using your desktop PC...

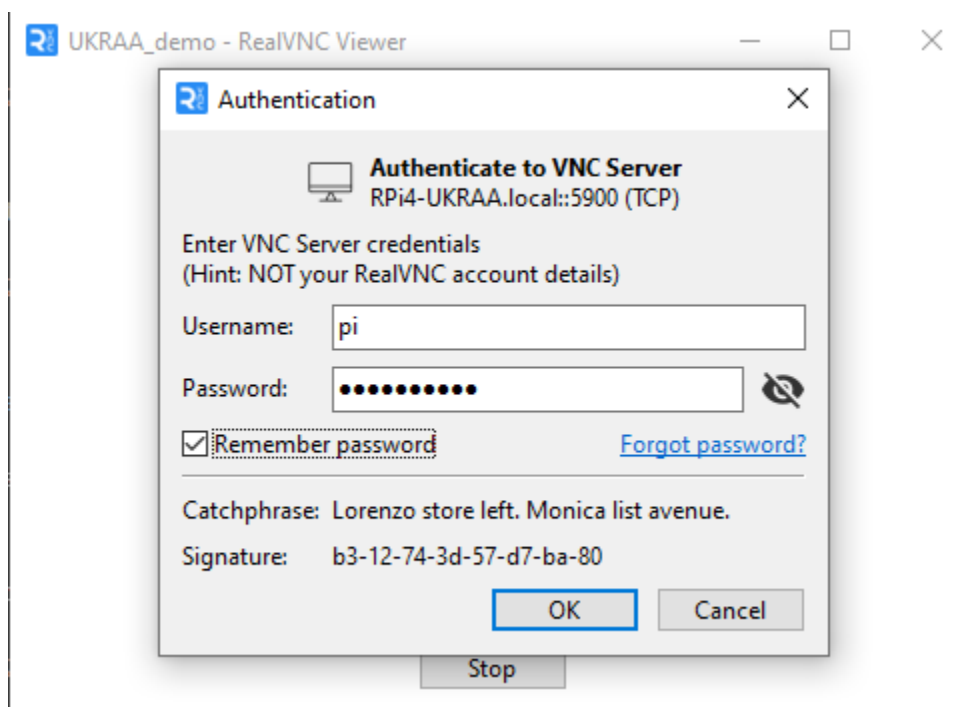
1. Create a **New connections** (CTRL-N) and enter the hostname into the **VNC Server:** box, add friendly name in the **Name** box and select **OK**.



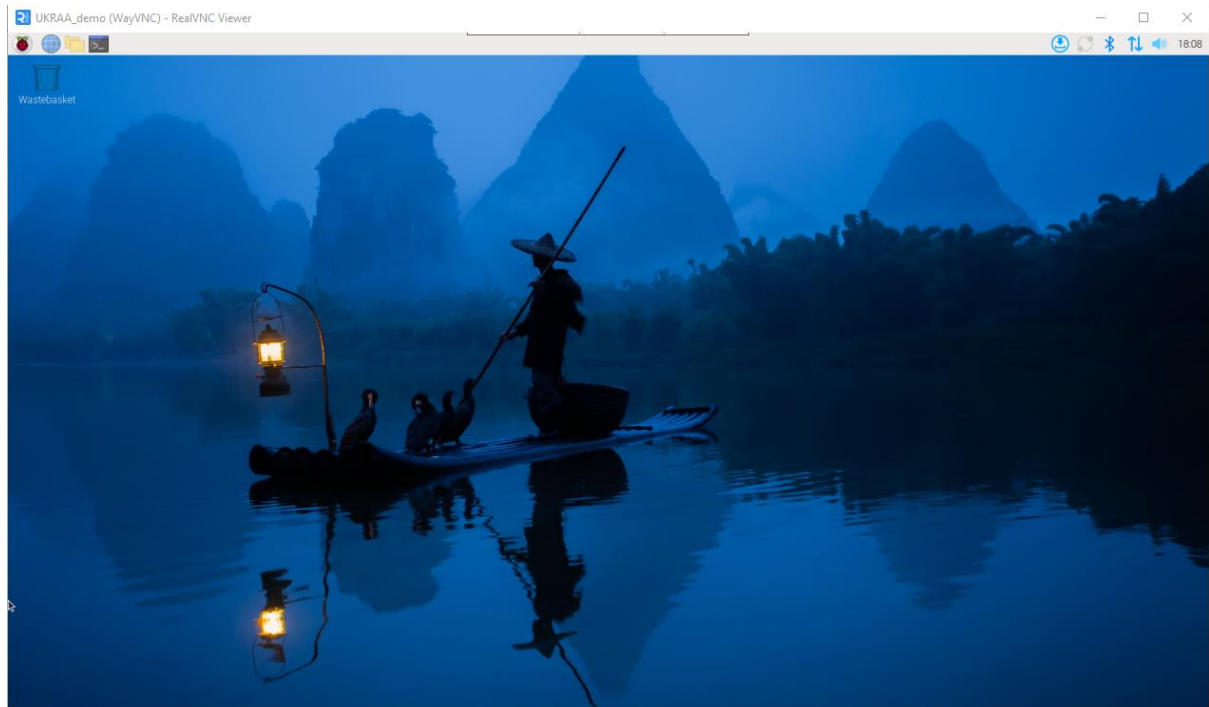
2. Open this VNC connection to your RPi; you will get a **VNC server not recognised** window from RealVNC. Select **Continue**



3. You will now get an **Authentication** window from RealVNC, enter your **Username** and **Password** details and select **Remember password** to make it easier to login in future.



4. You are now remotely accessing your RPi from your desktop PC...



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Contact us

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